1.7.4 Ratio scales:

Ratio scales are important when it comes to measurement scales because they tell us about the order, they tell us the exact value between units, and they also have an absolute zero–which allows for a wide range of both descriptive and inferential statistics to be applied. Good examples of ratio variables include height and weight. Ratio scales provide a wealth of possibilities when it comes to statistical analysis. These variables can be meaningfully added, subtracted, multiplied, divided). Central tendency can be measured by mean, median, or mode; Measures of dispersion, such as standard deviation and coefficient of variation can also be calculated from ratio scales.

In summary, nominal variables are used to "name," or label a series of values. Ordinal scales provide good information about the order of choices, such as in a customer satisfaction survey. Interval scales give us the order of values plus the ability to quantify the difference between each one. Finally, Ratio scales give us the ultimate–order, interval values, plus the ability to calculate ratios since a "true zero" can be defined. The distinction made here among nominal, ordinal, interval and ratio data are very much important as these concepts used in computers for solving statistical problems using statistical packages like SPSS, SAS, R, STATA etc.,

Points to Remember	
● Characteristics of statistics	Aggregate of facts collected in systematic manner for a specific purpose. Affected by large number of causes to marked extent. Numerically expressed. Enumerated or estimated with a reasonable degree of accuracy. To be placed in relation to the other.
● Functions of Statistics	Collection, Classification, Condensation, Comparison, Correlation, Causation, Chance
● Scope and Applications	Actuarial science, Commerce, Economics, Medicine, Agriculture, Industry, Information Technology Government
● Applications of Big Data	Smarter Healthcare, Retail, Traffic control Manufacturing, Search Quality, Sales promotion
● Types of Data	Quantitative Data, Qualitative Data
● Measurement Scales	Nominal scale, Ordinal scale, Interval scale, Ratio scale

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